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1. Ramesh's basic salary is input through the keyboard. His dearness allowance is 40% of basic salary, and house rent allowance is 20% of basic salary. Write a program to calculate his gross salary.

```
In [13]: basic_salary = float(input('enter your basic salary:'))

dearness_allowance = basic_salary * 40/100

house_rent_allowance = basic_salary * 20/100

Gross_Salary = (basic_salary + dearness_allowance + house_rent_allowance)

print(f'dearness_allowance: {dearness_allowance}, house_rent_allowance : {house_rent_allowance}')
print('Ramesh's Gross salary is:',Gross_Salary)
```

```
dearness_allowance: 180000.0, house_rent_allowance : 90000.0
Ramesh's Gross salary is: 720000.0
```

2. The distance between two cities (in km.) is input through the keyboard. Write a program to convert and print this distance in meters, feet, inches and centimeters.

```
In [12]: distance_km = float(input('Enter the distance between two cities in km:'))

meters = distance_km * 1000

feet = distance_km * 3280.84

inches = distance_km * 39370.1

centimeters = distance_km * 100000

print(f'the distance between two cities:, meters: {meters}, feet: {feet}, inches: {inches}, centimeters: {centimeters}')
```

```
the distance between two cities:, meters: 23000.0, feet: 75459.32, inches: 905512.2999999999, centimeters: 2300000.0
```

3. If the marks obtained by a student in five different subjects are input through the keyboard, find out the aggregate marks and percentage marks obtained by the student. Assume that the maximum marks that can be obtained by a student in each subject is 100.

```
In [16]: s1=float(input('marks obtained in subject 1:'))
s2=float(input('marks obtained in subject 2:'))
s3=float(input('marks obtained in subject 3:'))
s4=float(input('marks obtained in subject 4:'))
s5=float(input('marks obtained in subject 5:'))

Total_Maximum_Marks =500

Aggregate = s1 + s2 + s3 + s4 + s5
print('the total marks is:',Aggregate)

Percentage = (Aggregate / Total_Maximum_Marks) * 100
print('the percentage is:',Percentage)
```

the total marks is: 356.0

the percentage is: 71.2

4. Temperature of a city in Fahrenheit degrees is input through the keyboard. Write a program to convert this temperature into Centigrade degrees.

```
In [4]: fahrenheit = float(input('Enter temperature of a city in Fahrenheit: '))

centigrade = (fahrenheit - 32) * 5 / 9

print('Temperature of a city in Centigrade =', round(centigrade,2))
```

Temperature of a city in Centigrade = -4.44

5. The length & breadth of a rectangle and radius of a circle are input through the keyboard. Write a program to calculate the area & perimeter of the rectangle, and the area & circumference of the circle.

```
In [26]: length = float(input('Enter the length of rectangle :'))
breadth = float(input('Enter the breadth of rectangle :'))

r =float(input('Enter radius of a circle :'))
```

```

area_of_rectangle = length * breadth

Perimeter = 2*(length + breadth)

area_of_circle = 3.14 * r * r

circumference_of_circle = 2 * 3.14 * r

print(f'length : {length}, breadth : {breadth}, area_of_rectangle : {area_of_rectangle }, Perimeter : {Perimeter}')

print(f'radius of circle : {r} ,area_of_circle : {area_of_circle }, circumference_of_circle : {circumference_of_circle}')

```

length : 10.0, breadth : 9.0, area_of_rectangle : 90.0, Perimeter : 38.0
radius of circle : 34.0 ,area_of_circle : 3629.84, circumference_of_circle : 213.52

6. Two numbers are input through the keyboard into two locations C and D. Write a program to interchange the contents of C and D.

```

In [5]: C = float(input("Enter value for C: "))
        D = float(input("Enter value for D: "))

        # Interchange contents
        temp = C
        C = D
        D = temp

        print(f"After interchanging, C = {C} and D = {D}")

```

After interchanging, C = 20.0 and D = 10.0

7. If a five-digit number is input through the keyboard, write a program to reverse the number.

```

In [17]: number = int(input('Enter the five digit number:'))

        d1 = number // 10000
        d2 = (number // 1000) % 10
        d3 = (number // 100) % 10
        d4 = (number // 10) % 10
        d5 = number % 10

```

```
reverse = (d5 * 10000) + (d4 * 1000) + (d3 * 100) + (d2 * 10) + d1  
  
print('Reverse number =', reverse)
```

Reverse number = 36543

8. If a four-digit number is input through the keyboard, write a program to obtain the sum of the first and last digit of this number.

```
In [2]: number = int(input('Enter a four digit number : '))  
  
d1 = number // 1000  
d2 = (number // 100) % 10  
d3 = (number // 10) % 10  
d4 = number % 10  
  
sum = d1 + d4  
print('the sum is :',sum)
```

the sum is : 15

9. In a town, the percentage of men is 52. The percentage of total literacy is 48. If total percentage of literate men is 35 of the total population, write a program to find the total number of illiterate men and women if the population of the town is 80,000.

```
In [6]: total_population = 80000  
  
pct_men = 52  
pct_total_literacy = 48  
pct_lit_men = 35  
  
total_men = (pct_men / 100) * total_population  
total_women = total_population - total_men  
  
total_literate = (pct_total_literacy / 100) * total_population  
lit_men = (pct_lit_men / 100) * total_population  
  
illit_men = total_men - lit_men  
lit_women = total_literate - lit_men  
illit_women = total_women - lit_women
```

```
print(f"Total Population: {total_population}")
print(f"Total Number of Illiterate Men: {int(illit_men)}")
print(f"Total Number of Illiterate Women: {int(illit_women)}")
```

Total Population: 80000

Total Number of Illiterate Men: 13600

Total Number of Illiterate Women: 28000

10. A cashier has currency notes of denominations 10, 50 and 100. If the amount to be withdrawn is input through the keyboard in hundreds, find the total number of currency notes of each denomination the cashier will have to give to the withdrawer.

```
In [8]: amount = int(input("Enter the amount to be withdrawn : "))

notes_100 = amount // 100
remaining_amount = amount % 100

notes_50 = remaining_amount // 50
remaining_amount = remaining_amount % 50

notes_10 = remaining_amount // 10
remaining_amount = remaining_amount % 10

print(f"\nCurrency Breakdown for Rs.: {amount}")
print(f"Rs. 100 notes: {notes_100}")
print(f"Rs. 50 notes : {notes_50}")
print(f"Rs. 10 notes : {notes_10}")

if remaining_amount > 0:
    print(f"Remaining change that cannot be given in available denominations: Rs. {remaining_amount}")
```

Currency Breakdown for Rs.: 234000

Rs. 100 notes: 23400

Rs. 50 notes : 0

Rs. 10 notes : 0

In []: